Corrosion

Corrosion is the deterioration of metal due to a reaction with the environment. Although it generally causes detrimental issues, not all corrosions are negative, for example a formation of a layer that protects the metal from the environ

Corrosion occurs due to metals wanting to reach a more stable state, as the metallic state is unstable and can reach better stability via corrosion. Usually, it forms a metallic oxide or compound in the stable state.

In the case of oxides, some metals do not oxidise readily and thus will not form an oxide layer naturally, like Au and other precious metals. Other metals form volatile oxides at high temperatures, like Mo oxides, which vaporise at temperatures in excess of 1100C. Other oxides act as a protective coating, like Al and Cr.

Generally, there are two corrosion types,

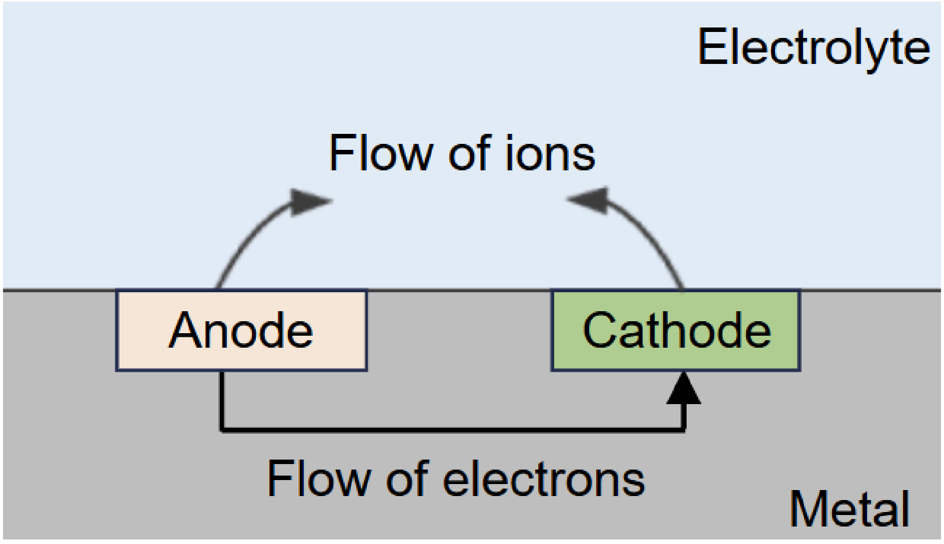
Chemical, when the metal is attacked by a corrosive medium

Electrochemical, when the metal atoms lose electrons and oxidise. Since the most common type of corrosion is via this mode, we will focus on this type of corrosion.

Electrochemical Corrosion

- There are 4 conditions for corrosion to take place:
- There must be differences in the metal's surface, forming
  - anodes/anodic sites
  - cathodes/cathodic sites
- An electrolyte in contact with the metal
- A conductive path for electrons flow between electrodes
- The four conditions form a corrosion cell, where the electrolyte is the corroding medium, and the anode is where the metal is corroded as it loses/produces electrons, while the cathode is not consumed and is protected during corrosion.
- Sine the anode loses electrons, it oxidises and the electrons flow to the cathode, which gains electrons and thus undergoes reduction. The flow of

electrons happens in the metal, while ions from the anode flow towards the cathode in the electrolyte.



At the anode:

Corroding metal enters the electrolyte as ions, and electrons flow to the cathode in the metal. Thus, two currents occur from the anode to the cathode. The oxidisation half equation for a metal of valency 2 is

$$M \rightarrow M^2 + 2e^-$$

At the cathode:

Reduction must occur to balance out the oxidisation, by reacting with the water and oxygen in the electrolyte to form hydroxide ions via the electrons.

$$2H_2O + O_2 + 4e^- \rightarrow 4OH^-$$

Thus, both reactions form ions in the electrolyte, which combine as

$$M^2 + 2OH^- \rightarrow M(OH)_2$$

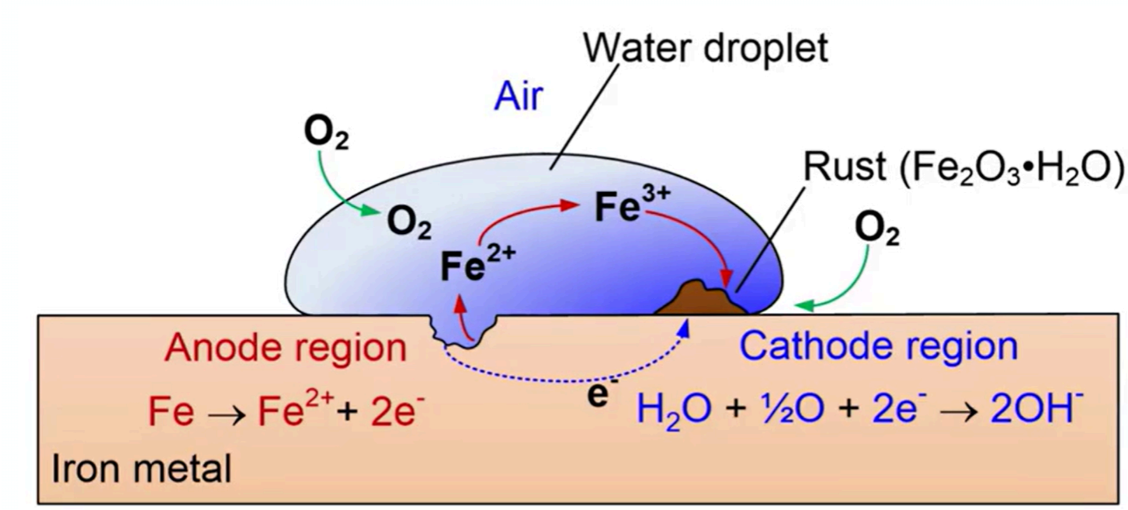
For steel, After Fe(OH)2 is formed from the first corrosion process, it can continue to further oxidise from Fe^2+ to Fe^3+ and form Fe(OH)3:

$$Fe + \frac{1}{2}O_2 + H_2O \rightarrow Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$$

$Fe(OH)_2$ is further oxidised to form $Fe(OH)_3$

$$2Fe(OH)_2 + \frac{1}{2}O_2 + H_2O \rightarrow 2Fe(OH)_3$$

The further oxidisation to Fe(OH)3 causes Fe3+ ions to be present in the electrolyte, which deposits on the metal as Fe2O3, or rust.

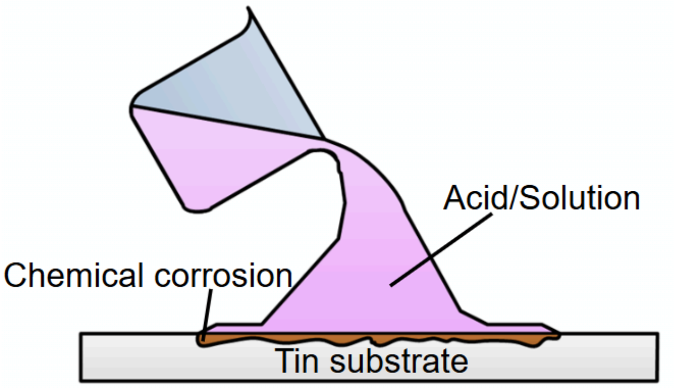


The corrosion of the metal in the atmosphere is affected by the moisture, dust and pollutants in the air (water and air pollution). The presence of moisture allows for the electrolyte to be directly in contact with the electrodes to cause further corrosion. Dust also acts as a carrier to allow water to be in contact with the metal, and pollutants like SO2 and H2S are acidic are incite further reaction.

Corrosion Locations

There are generally two classes of corrosion,

Uniform corrosion, which is the most common but least damaging, as it is usually due to direct attack via chemical media like acids and solvents. The corrosion is not centralised and happens across the entire surface.



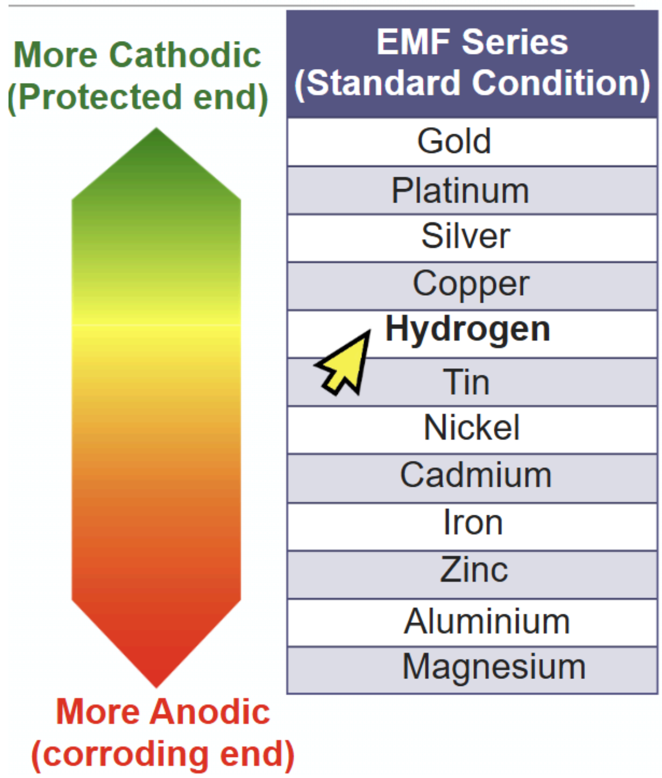
Every other form of corrosion is called localised corrosion.

It occurs at specific locations on the metal and is most concentrated at specific locations, but can spread quickly and can make steels brittle.

There are 6 types of corrosion that fit under localised corrosion:

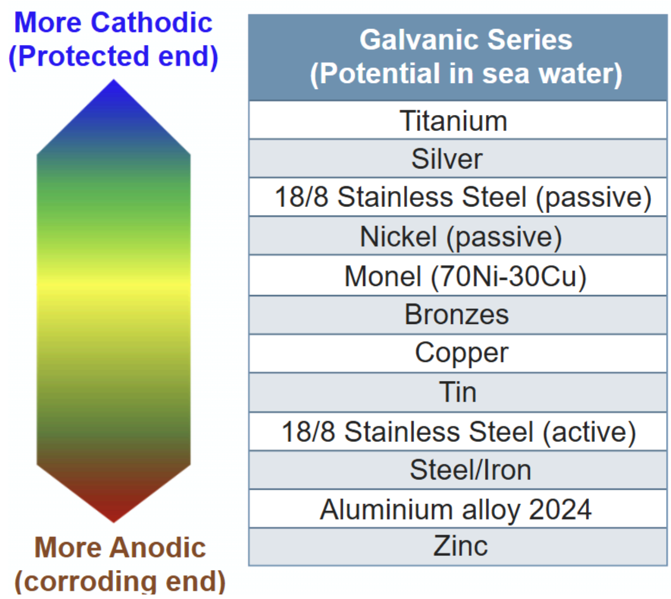
Galvanic Corrosion

> Galvanic corrosion occurs when two different metals are in contact. There has to be an electrolyte and an electrical connection between the two. This causes galvanic corrosion to occur at the interface on the anode terminal.
>
> The anodic/cathodic behavior of the metal is determined on the electromotive force series.



Hydrogen is on the series as a base point, and all the metals can be measured against it to see how anodic or cathodic they are. The metals above hydrogen are more cathodic, and are thus more resistant to corrosion and are inert, and vice versa below hydrogen. The more cathodic it is, the higher the emf (+emf) and vice versa

There is also the galvanic series, which measures the emf with sea water as an electrolyte, and is more commonly used.

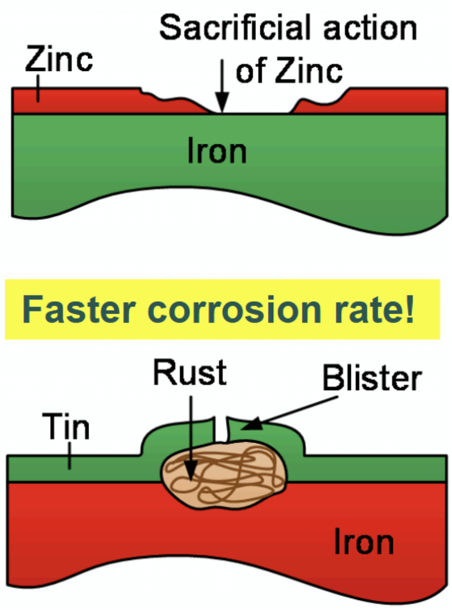


It follows a similar system, in that the higher up it is on the list, the more cathodic it is and the more resistant it is to corrosion.

The two main factors affection the rate of galvanic corrosion are:

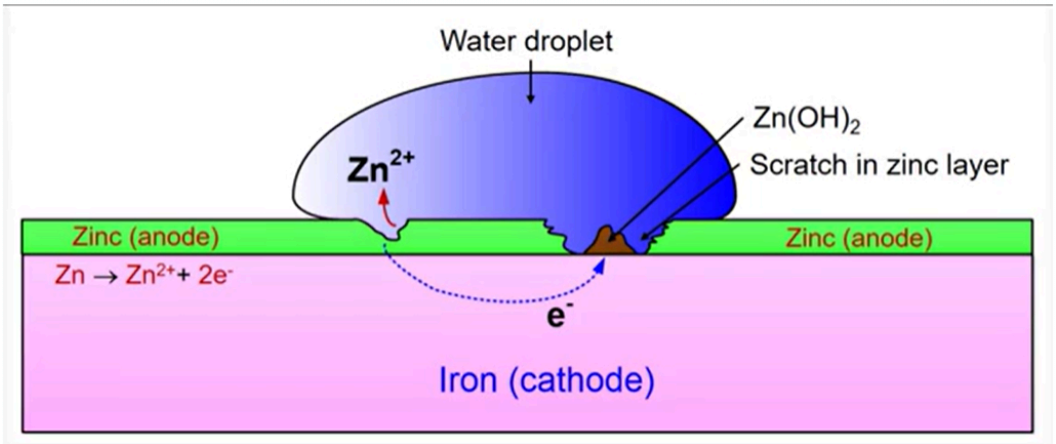
The difference in electrical potential, which is how far apart they are on the series, and a higher series causes higher corrosion rate,

The contact area ratio of the anode, where if the anode has a larger area, the corrosion will be more uniform and less deep, but a smaller anode compared to the cathode causes more concentrated and deeper corrosion, as well as a faster corrosion rate.



For example, notice the width and depth of the metal pairs, where the top example shows zinc as the anode, and the bottom example shows iron as the anode.

The larger zinc area compared to iron causes corrosion to be more widespread, but go less deep into the metal, while for the iron in contact with the tin, the scratch in the iron causes a concentrated area and thus a concentrated region of corrosion, which penetrates deeper into the metal, causing rust to build up faster too.



For example, for a piece of iron galvanised with zinc, a layer of zinc is formed on top of the iron. Despite the protective ZnO layer on the zinc, if the layer is scratched and an electrolyte like water is present, the zinc will act as the anode the iron will be the cathode, as zinc is more reactive than iron. This causes Zn2+ ions to be released into the electrolyte while electrons flow from the zinc into the iron cathode, which allows OH- to be formed at the scratch. As the scratch usually has a small area, it causes a more concentrated corrosion to occur, which allows more Zn(OH)2 to form at the scratch.

Crevice Corrosion

Crevice corrosion occurs under a stagnant electrolyte solution in a crevice like narrows openings, due to a limited amount of oxygen in the crevice.

Typically, these crevices are mechanical joints, like gaskets, washers, fastener heads, and also surface deposits, threads and lap joints

A crevice is generally any small gap, that in this case still allows moisture to seep through for corrosion to still occur. As the crevice is a small gap, it makes it harder for oxygen to get into the crevice, causing a low concentration of oxygen in the crevice and a higher oxygen concentration outside. The lower concentration of oxygen forms an anodic site while the higher oxygen concentration outside is now a cathodic site, forming a corrosion cell.

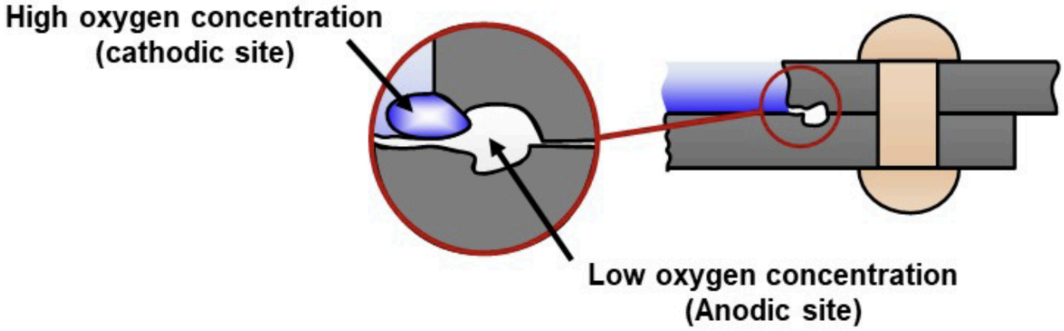
Recall the half equations for corrosion, being:

$$\text{Anodic Site: } M \to M^{2+} + 2e^-$$
$$\text{Cathodic Site: } 2H_2O + O_2 + 4e^- \to 4OH^-$$

Notice how oxygen is one of the reactants in the reduction equation. When an area has high concentrations of oxygen, it is more likely to undergo the cathodic reaction as the higher oxygen encourages the cathodic reaction, thus it will form a cathodic site.

In limited oxygen concentrations however, not enough oxygen is present to be able to undergo the cathodic reaction. Due to the cathodic reaction outside, the lower oxygen concentration region must respond with an anodic reaction to maintain redox, thus the crevice with low oxygen concentration becomes the anodic site.

This causes metal ions formed at the anodic crevice to migrate outwards and form compounds with the hydroxide at the cathodic site, forming corrosion deposits which can further limit the oxygen flow and further lower the oxygen concentration.



Thus, some ways of reducing crevice corrosion are:

- Using welded joints instead of bolted/riveted joints
- using non-absorbing gaskets (to fill the gaps and prevent water from reaching the metal)
- removing accumulated deposits
- designing containment vessels to avoid stagnant areas

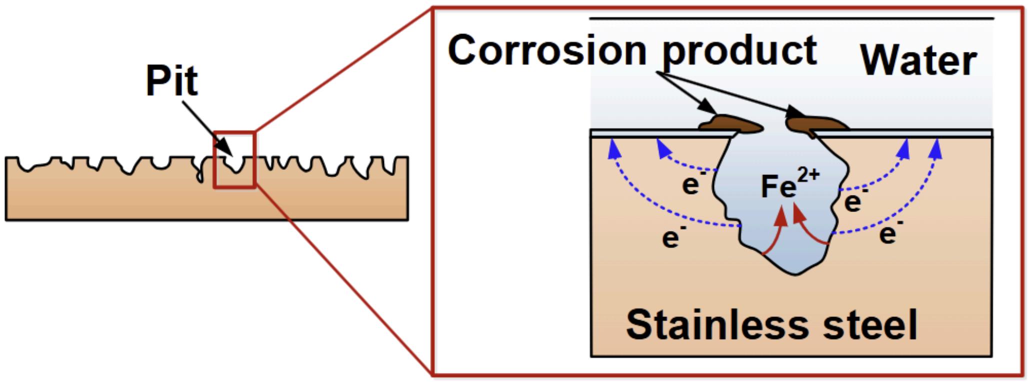
Pitting Corrosion

Similar to crevice corrosion, pitting corrosion occurs at holes or pits of the metal; heterogenous surfaces on the metal.

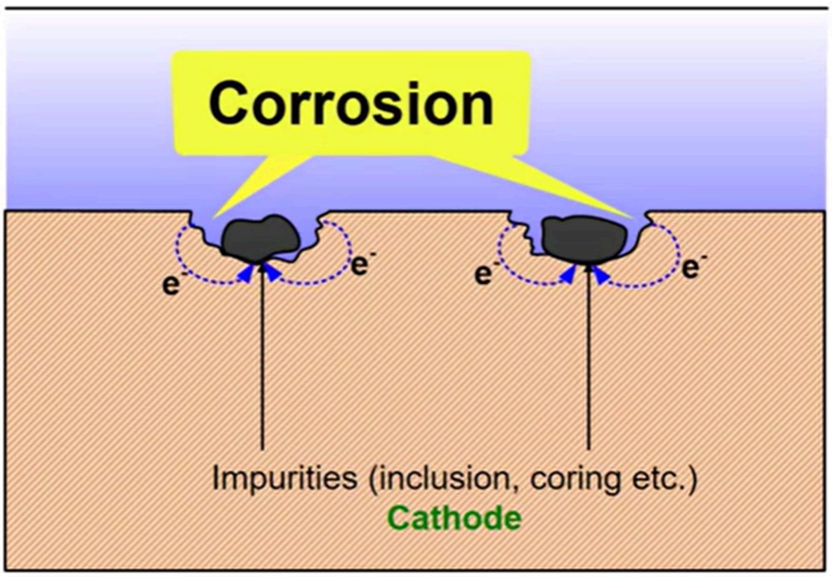
This is one of the more detrimental and dangerous types of corrosion as it is very destructive to structures and forms perforations in the metal.

The corrosion process is initiated by disruptions in the oxide film, likely due to an acidic environment, a low dissolved oxygen concentration or a high chlorine concentration. This causes the metal to be more anodic in contrast to the surface. Anodic sites can also be formed due to localised changes in the electrolyte at the different heterogeneities in the metal.

This allows corrosion to occur and it deposits corrosion compounds on the sides/top of the pits and holes, which may seal the holes. Not only does it make these corrosion sites hard to spot, it also forms crevices which can cause further crevice corrosion, causing further deterioration of the metal.



Similarly, heterogeneities like inclusions can act as cathodes compared to the base metal, encouraging corrosion similarly, where the more anodic surrounding metal deposits metal ions and corrodes.



As pitting corrosion continues and the pit continued to get corroded, the pits get deeper and causes further oxygen depletion in the deeper pit, which allows further corrosion to continue.

Pitting corrosion can be accelerated by increasing the:

- electrolyte acidity
- low dissolved oxygen concentration of electrolyte (decreasing oxygen concentration)
- chlorine ion concentration in electrolyte.

and can also be prevented by doing the opposite of these.

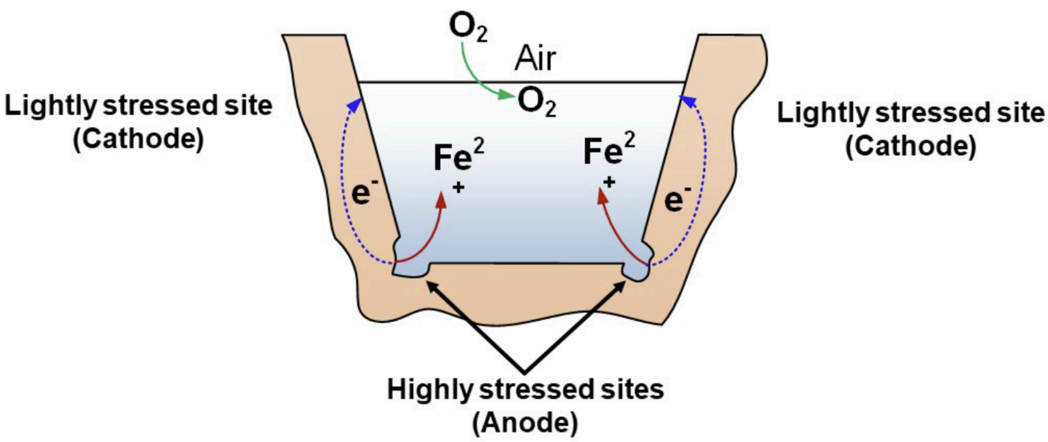
Stress Corrosion

Stress corrosion is due to both a corrosive environment and tensile stresses applied to the metal.

Stress concentrations like notches and grooves cause a stress gradient to be setup in the metal, which causes a potential difference due to the different stress levels. This causes regions of high stress levels to be more anodic and lower stress regions to be more cathodic, with accelerates the corrosion due to a corrosive environment.

This is most susceptible to cold worked parts due to the stresses present in the metal, further worsening ductility and other mechanical properties.

Especially with the corrosive environment, stress corrosion occurs, as well as hairline cracks at the corrosion sites, with little corrosion deposits. This form of cracking is also called stress corrosion cracking.



Stress corrosion can be prevented by:

removing stress concentration sites by changing the design

removing stresses by heat treatment like annealing

removing the corrosive environment.

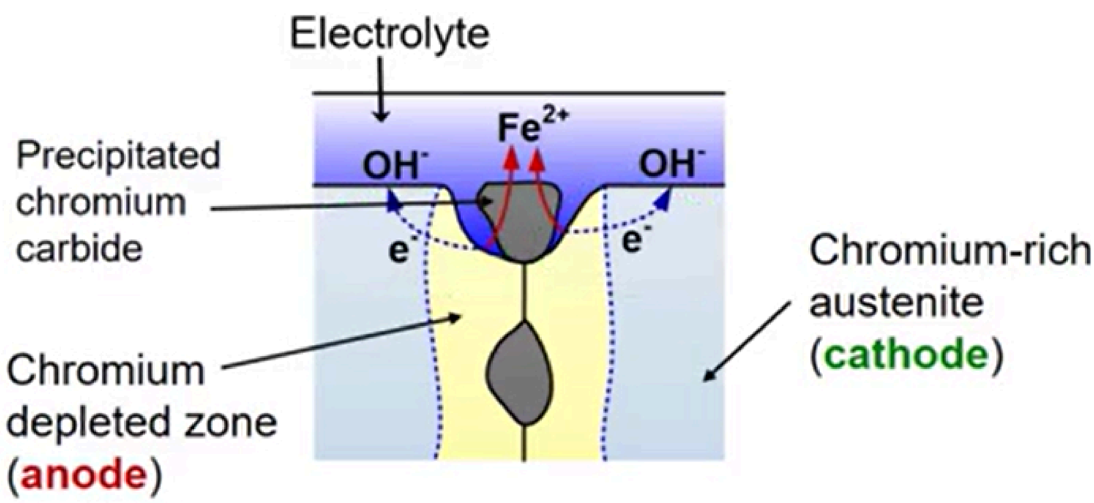
Intergranular Corrosion

Intergranular corrosion occurs at the grain boundaries in metals.

Usually, this corrosion is due to precipitates, which form and concentrate at the grain boundaries. As the precipitate forms at the grain boundaries, it causes the grain boundary to be more anodic, encouraging corrosion to occur.

The most common example is the intergranular corrosion of austenitic stainless steels (steels with chromium).

When these steels are heated to 500-800C and slowly cooled, it causes chromium carbide precipitates to form on the grain boundaries. This causes a chromium depleted zone adjacent to the grain boundary, which ends up as an anode. As the area near the grains is now more anodic, the grain boundary starts to corrode, by depositing metal ions into the electrolyte and causing electrons to flow into the chromium richer grains; the cathode.



This specific kind of corrosion is called weld decay. where welding causes the heating and cooling of the metal.

To prevent this, we can prevent the formation of chromium carbide precipitates, via:

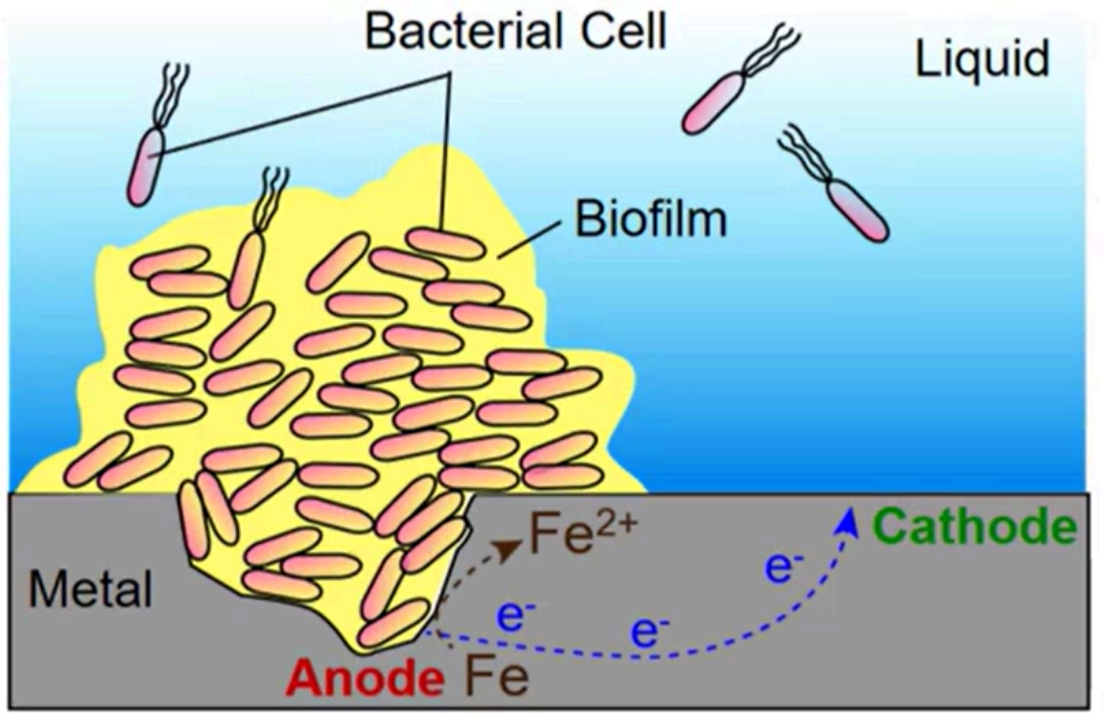
- high temp heat treatment to dissolve chromium carbide particles
- very rapid cooling in that temperature range
- reducing to <0.03 % wt of C to prevent chromium carbide from even forming
- adding carbide formers like Nb or Ti to prevent C from combining with Cr.

For cold worked aluminium alloys, they are more susceptible to a type of intergranular type of corrosion, called exfoliation, which causes expansion and hence bending stresses.

Microbiological Corrosion

Microbiological Corrosion is a type of corrosion due to microorganisms, like fungi and bacteria on the metal.

As microorganism colonies cluster around different sections of the metal, they consume oxygen around the region the colonies are located. This causes a lower oxygen concentration which causes the area of metal around the colony to be more anodic, and causes it to corrode. Unaffected areas of the metal act as cathodes and receive electrons from the anodic site. These microbes tend to form colonies due to adequate nutrients and corrosive by products, and can be mitigated by reducing these.



The specific strategies to reduce microbiological corrosion are

- modifying the surface to inhibit the growth of microorganisms, like a protective layer,
- control environmental factors like ph to discourage microbial growth
- adding chemical inhibitors like biocides.

To avoid corrosion, there are 5 general methods to properly mitigate corrosion.

Material Choice

This method involves using a metal that is already resistive to the corrosive environment.

It mainly involves adding alloying elements to steel to improve its corrosion resistance:

Cu; improves resistance to atmosphere, acidic and salt solutions

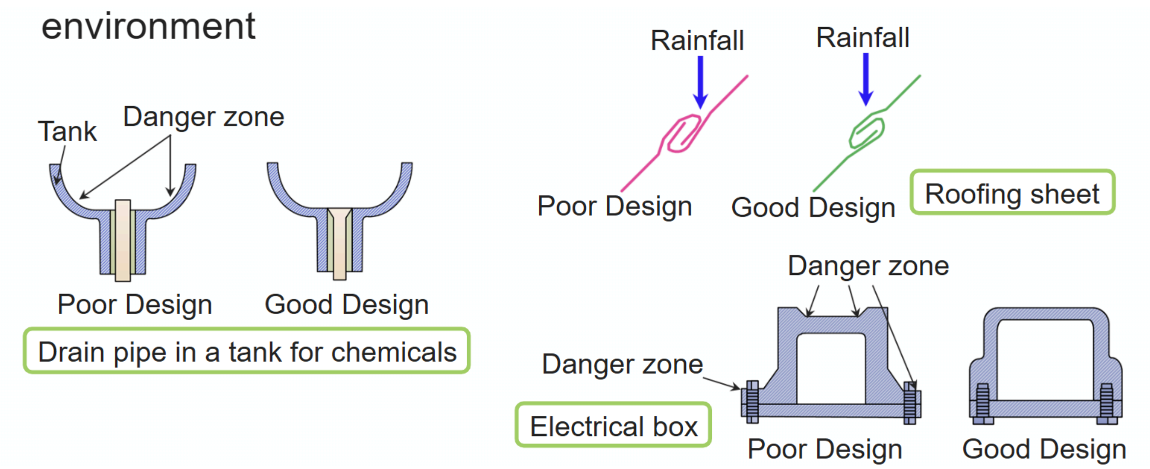
Cr, Al, Si; allows protective oxide film to form

Ti, Nb; forms carbides which prevents intergranular corrosion/weld decay

Mo; increases resistance to pitting and crevice corrosion

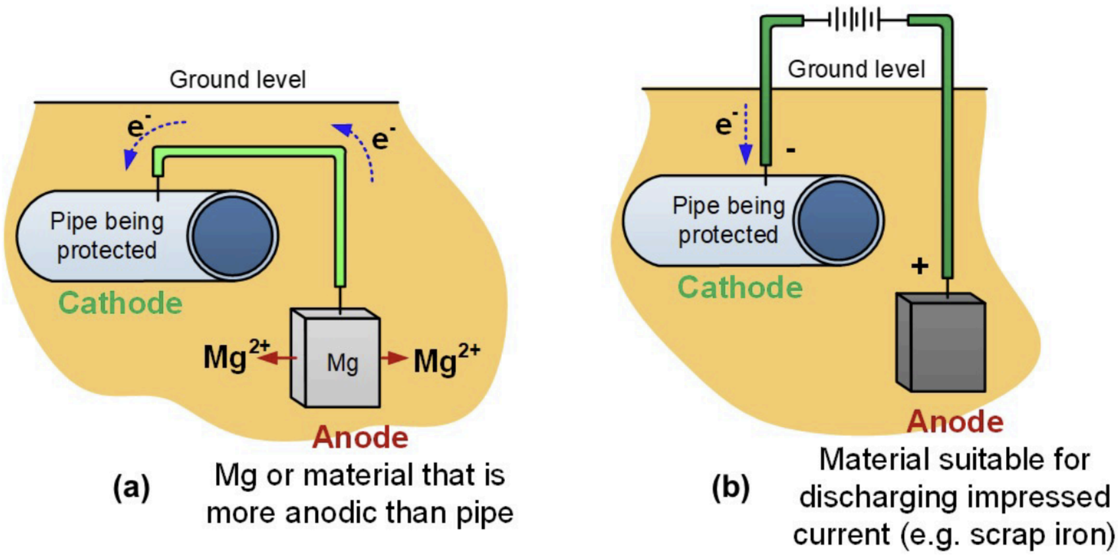
Preventive Design

This method involves minimise contact of the metal with the service environment. Generally, this involves preventing the build up of liquids on the part, like orienting joints to prevent retained rainwater, or designing vessels to ensure all the retained rainwater is drained out properly.



Cathodic Prevention

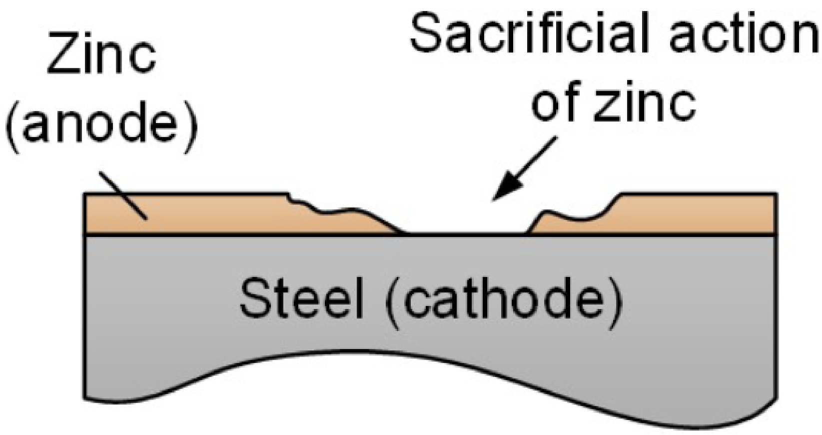
Cathodic prevention involves directing electrons to the metal, causing it to act as a cathode. The two main ways to direct electron flow to the metal is with a more reactive metal to act as an anode, or with a voltage with an electrical connection from the metal to the scrap piece of metal.



In a corrosive environment, when the metal is connected to a more reactive metal, causing it to corrode and become anodic, which loses electrons and protects the metal as a cathode. Using a voltage also provides electron flow to make the metal more cathodic, protecting it.

During this, the anode corrodes, and electrons are supplied to the protected metal, preventing the metal from undergoing a anodic reaction. Hence, since the anode is consumed in this process, it needs to be replaced. Typically, Zn or Mg is used as the anode.

Another cathodic prevention method is galvanisation, where a metal like steel is coated with zinc to act as a protective barrier as well as an anode for the steel to undergo cathodic protection, when the protective zinc layer is disrupted.



This method of corrosion prevention is used in protecting storage tanks, protecting steels in ships, and steel components in contact with a slurry.

Protective Coating

There are 3 main classes of coatings that are used to protect metals,

Metallic

there are 3 methods to apply a metal coating:

Electroplating, by setting up an electrolytic cell to have metal ions from one electrode deposit on the metal, to form a metallic coating

Coating in liquid state, by immersing or coating the metal with a molten metal, like hot dip galvanising, tinning, metal spraying and thermal spraying

Cladding, by covering a metal with other metal sheets, like cladding pure Al on duralumin, or sandwiching mild steel between layers of stainless steel.

Inorganic

The two main inorganic coatings are ceramic and glass, for withstanding different environments.

Ceramic coatings are used to withstand high temperature environments, like for engine blocks or tubing. Ceramic coatings also provide a smooth finish to the metal.

Glass coatings are used to withstand chemically corrosive environments, like for heat exchangers.

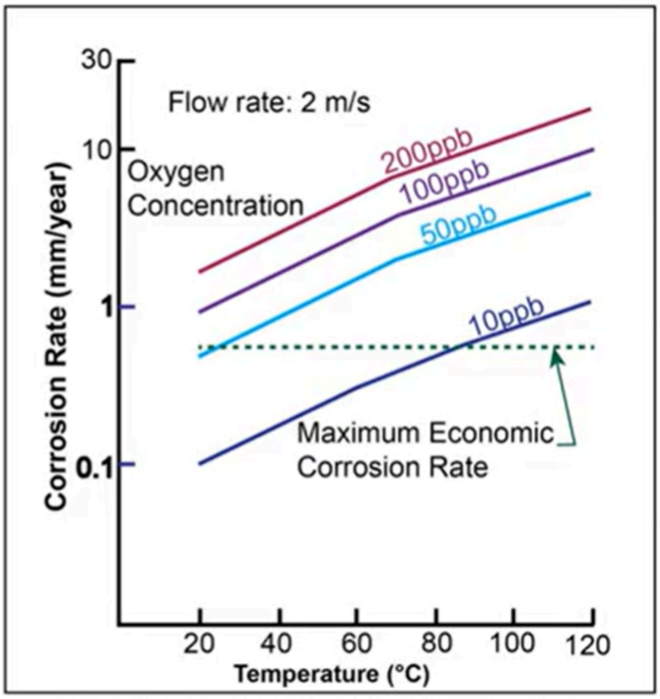
Organic

Some common coatings are paint, lacquer and varnish, which provide a thin, yet tough and durable barrier. Some examples include painted fuselages, organic coated fasteners, epoxy coated medical components and PTFE coated pipelines.

Environmental Control

Environmental control involves changing the environment to reduce its corrosive effect on metals. The five main parameters to reduce the corrosive effect are

Temperature Reduction



Ref: J. Oldfield and B. Todd Corrosion consideration in selecting metals for flash chambers, Desalination, 31, 1979, pp.365-383

From the chart, for a given flow rate, as temperature increases, the corrosion rate will increase, thus corrosion can be reduced by reducing the corrosion rate.

Oxygen Concentration Reduction

Likewise from the chart, as the oxygen concentration in parts per billion increases, the corrosion rate will also increase too, and should be reduced by doing the opposite.

Hydrogen Ion Concentration reduction

Other than an oxygen concentration difference, crevice corrosion may also be caused by the difference in concentration of hydrogen ions. Typically, hydrogen ion concentration is due to stagnant solutions, same as the conditions for crevice corrosion to take place. Thus, to reduce ion concentration, agitation of the solution can mitigate this.

Fluid Velocity reduction

Metals in direct contact to fast flowing fluids are more likely to be subjected to erosion corrosion, and thus reducing the fluid velocity reduces the risk.

Electrolyte Inhibitors

Inhibitors work by slowing down either the anodic or cathodic reaction, like via forming thin protective films, forming bulky precipitates to coat the metal or by combining with the corrosion deposits to form a protective layer.

Some examples include sprays with galvanic rich material to undergo cold galvanisation, or volatile corrosive inhibitors which deposit on the metal surface.